

Syed Hussam Abbas

Plant Molecular Biologist | RNA Surveillance, Stress Signalling and Heat Stress Memory



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RESEARCH INTERESTS

Post-transcriptional regulation of gene expression in plants, with a focus on nonsense-mediated mRNA decay (NMD), alternative splicing, heat shock factor (HSF) biology, and thermomemory in *Arabidopsis thaliana*; building on earlier work on transcription fidelity and alternative splicing control. Broader interests encompass RNA surveillance mechanisms, transcriptome and translome plasticity under abiotic stress, and the interplay between post-transcriptional and translational quality control; including how RNA surveillance is coordinated with protein and organelle turnover, such as selective autophagy and senescence regulators including ORE1, during recovery from heat stress.

EDUCATION

PhD in Plant Sciences

Oct 2022 – Expected 2026

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

Supervisor: Dr. Tibor Csorba, Plant Stress Biology Group

Thesis: “Post-transcriptional regulation of heat stress response and thermomemory by nonsense-mediated decay in *Arabidopsis thaliana*”

- Primary focus: NMD-mediated regulation of HSFA2 isoform balance during heat stress acclimation and thermomemory in *Arabidopsis*
- Contributing work: transcriptional fidelity and co-transcriptional splicing control under heat stress (TFIIS/NMD cooperative pathway)
- Manuscript submitted to Journal of Experimental Botany as first author; co-authored work published in The Plant Cell (2025) and The Plant Journal (2025)

MSc in Plant Breeding and Genetics

2019 – 2022

University of Agriculture Faisalabad, Pakistan

Thesis: “Assessment of association among different yield contributing factors in maize (*Zea mays* L.) germplasm”

RESEARCH EXPERIENCE

Doctoral Researcher

Oct 2022 – Present

Plant Stress Biology Group, MATE University, Gödöllő, Hungary

- Investigated NMD pathway regulation of HSFA2 alternative splicing isoforms (HSFA2-I and HSFA2-II) during heat stress acclimation and thermomemory using *upf* mutant lines
- Designed and executed controlled heat stress, acclimation, and thermomemory phenotyping assays across multiple *Arabidopsis* genotypes
- Developed and optimized heat stress phenotyping regimes, establishing standardized acclimation, recovery and memory-scoring protocols for multiple *Arabidopsis* genotypes
- Performed isoform-specific RT-qPCR to dissect temperature-dependent splicing outcomes and NMD-sensitive transcript dynamics
- Conducted western blotting and SDS-PAGE for protein-level validation of stress response phenotypes

- Applied FPLC-based protein purification for biochemical characterization of stress-related protein complexes
- Contributed RT-qPCR, genotyping, and western blotting to a co-authored study on TFIIS/NMD cooperative transcriptome quality control under heat stress (Szaker et al., *Plant Cell*, 2025)
- Performed MPA treatment experiments and confocal/fluorescence microscopy for cellular phenotyping
- Conducted bioinformatics analyses including RNA-seq data interpretation and multi-omics dataset integration

Research Assistant

Sep 2020 – Apr 2022

University of Agriculture Faisalabad, Pakistan

- Supported molecular biology experiments including nucleic acid extraction, PCR-based genotyping, and sample preparation
- Contributed to data collection, experimental execution, and results interpretation across active research projects

MSc Thesis Research

Sep 2021 – May 2022

University of Agriculture Faisalabad, Pakistan

- Conducted independent research in plant breeding and genetics; designed experiments, executed molecular analyses, and completed thesis

Research Intern

Feb 2019 – May 2019

Ayub Agricultural Research Institute, Faisalabad, Pakistan

- Assisted in laboratory and field-based experimental work; supported data recording and research activities

PUBLICATIONS

1. **Abbas S. H.**, Darkó É., Erős T. and Csorba T. (2026). Nonsense-mediated decay promotes heat stress recovery independently of HsfA2/HsfA3 in *Arabidopsis*. *Manuscript submitted to Journal of Experimental Botany*.
2. Szaker H. M.*, Verma R.*, Szádeczky-Kardoss I., Gál N., **Abbas S. H.**, Darkó É., Pettkó-Szandtner A., Silhavy D. and Csorba T. (2025). Transcription fidelity and control of alternative splicing contribute to heat stress survival in *Arabidopsis*. *The Plant Cell*, 37(11): koaf256. <https://doi.org/10.1093/plcell/koaf256>
3. Nyikó T., Gyula P., Ráth S., Sós-Hegedűs A., Csorba T., **Abbas S. H.**, Bóka K., Pettkó-Szandtner A., Móricz Á. M., Molnár B. P., Erdei A. L. and Szittya G. (2025). INCREASED DNA METHYLATION 3 forms a potential chromatin remodelling complex with HAIRPLUS to regulate DNA methylation and trichome development in tomato. *The Plant Journal*, 121: e70085. <https://doi.org/10.1111/tpj.70085>
4. Singh K., **Abbas S. H.**, Singh S., Iqbal N., Kocsy G., Pérez-Pérez J. M., Riyazuddin R. (2025). Deciphering the Plants' Response to Flood-Induced Hypoxia and Anoxia at Molecular Level. In: Riyazuddin R., Gupta R., Ramteke P. W., Sayyed R. (eds) *Plant Flooding. Plant in Challenging Environments*. Springer, Cham, 6:93-112. https://doi.org/10.1007/978-3-031-83068-6_5

TECHNICAL SKILLS

RNA & Molecular Biology

- RT-qPCR / isoform-specific qPCR, PCR, molecular cloning, DNA/RNA extraction, plasmid preparation
- Alternative splicing analysis, NMD-sensitive transcript quantification
- MPA treatment, heat stress acclimation and thermomemory assay design

Protein Biochemistry

- Western blotting, SDS-PAGE, co-immunoprecipitation (Co-IP)
- Protein purification by FPLC; biochemical characterization of protein complexes

Microscopy & Imaging

- Confocal microscopy, fluorescence microscopy, optical microscopy

Bioinformatics & Data Analysis

- RNA-seq data interpretation, differential expression and splicing analysis
- R programming, statistical analysis, data visualization
- Multi-omics dataset integration; familiarity with plant genome databases (TAIR, Ensembl Plants)

Plant Biology

- Arabidopsis thaliana cultivation, transformation, and phenotyping
- Controlled abiotic stress experiments (heat, hypoxia/anoxia)
- Genotyping of T-DNA insertion and EMS mutant lines

ADDITIONAL EXPERIENCE

Campus Mentor

Jul 2024 – Present

Stipendium Hungaricum Mentor Network (HOOK), Hungary

- Mentored international PhD and MSc students on academic integration, administrative procedures, and well-being
- Coordinated student activities and prepared structured monthly progress reports

R&D Associate

Jun 2022 – Sep 2022

BioTech Life Sciences Pvt Ltd, Multan, Pakistan

- Assisted in laboratory experiments and product-oriented R&D; supported data documentation and analysis

AWARDS & SCHOLARSHIPS

- SH Dissertation Scholarship (Hungarian government scholarship for international doctoral researchers)
- Stipendium Hungaricum Scholarship (Hungarian government scholarship for international doctoral researchers)
- Pannónia Mobility Scholarship (Hungarian mobility award for international research activities)
- Chief Minister's Merit Scholarship (competitive academic merit award, Pakistan)
- Prime Minister's Laptop Award (national academic excellence award, Pakistan)

REFERENCES

Available upon request.